Optimal Trading Filters: a Wiener–Hopf Approach

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Keywords. Optimal trading; market impact; propagator model; inventory risk; Wiener–Hopf factorization; linear filters; fractional calculus; adapted stochastic control.
JEL classification. C61; G11; G12; G14.
MSC classification. 45E10; 60G35; 91G80; 26A33.

Abstract

A trader holding a predictive signal must weigh the expected gain against the cost of acquiring the position and the risk of holding it. Under the propagator model of market impact, trading-induced price changes involve the entire history of past trades. Existing treatments of the resulting path-wise quadratic program characterize the optimum implicitly, as the solution of an integral equation or an equivalent forward–backward system. In this paper I show how stationary solutions can be computed analytically from the signal and its forecasts using a Wiener-Hopf factorization of the friction operator. For rational frictions — inventory risk, temporary cost, and exponential resilience — the trading filter reduces to finitely many exponential moving averages, and we recover the Neuman–Voß solution away from horizon boundaries. In the (empirically more realistic) case of scale-free impact kernels, the optimal trading filter decays algebraically rather than exponentially at large frequencies; in the limit of small risk aversion, the target position is a fractional integral of the signal. Unlike its exponential (Gârleanu–Pedersen) cousin, this solution never surfs its own impact or generates block trades.

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